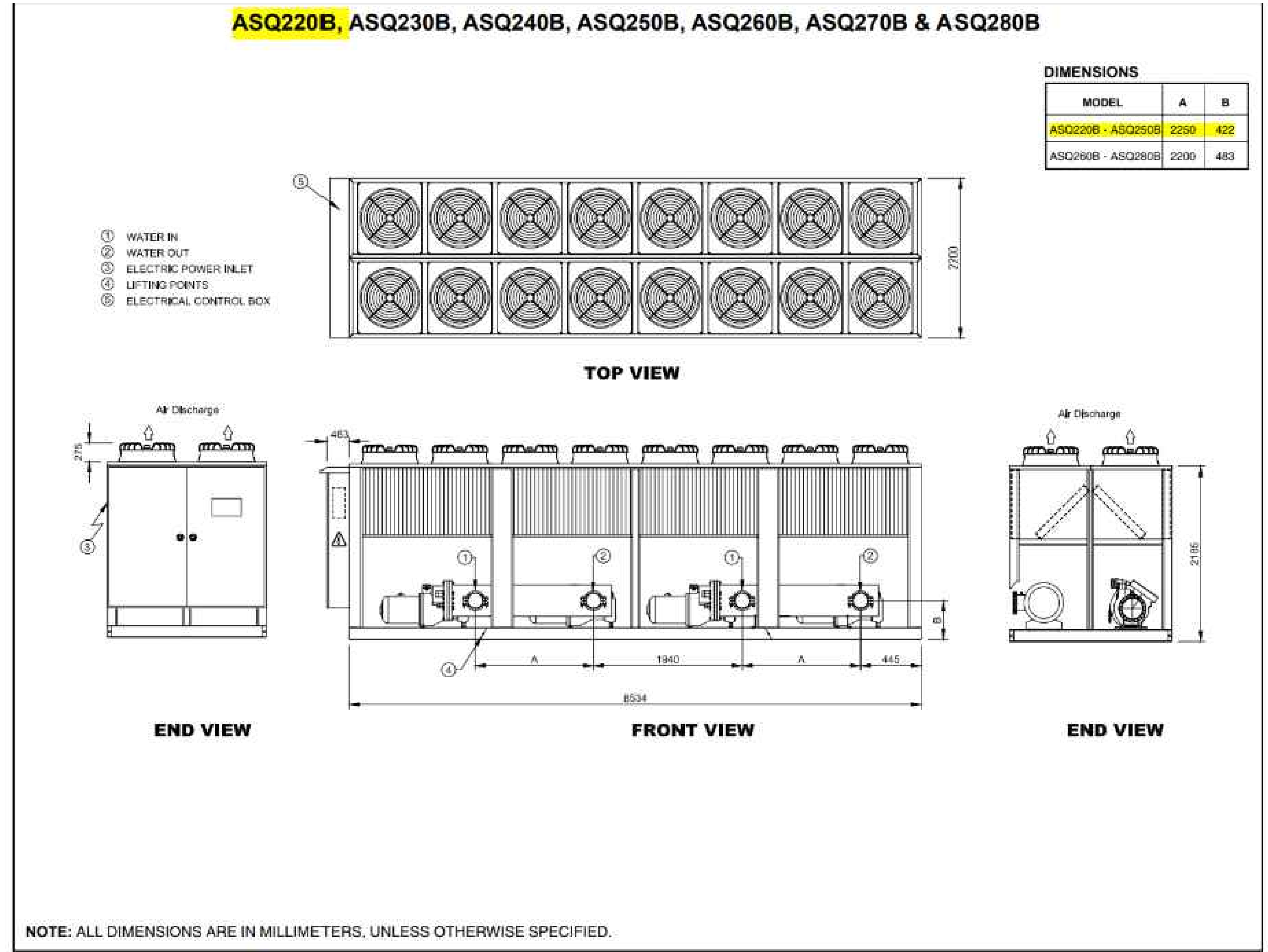
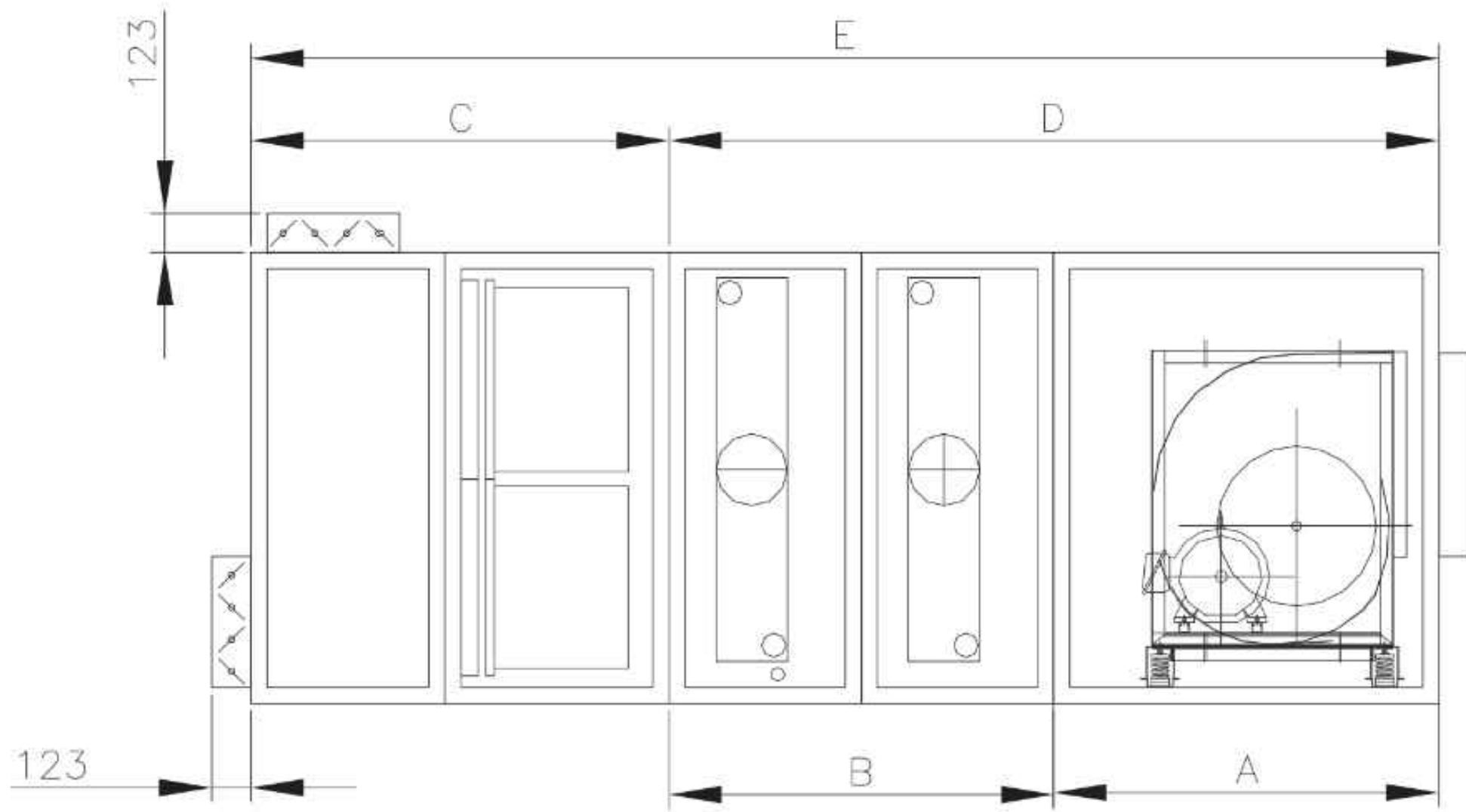


Chiller Schedule						
Manufacturer	Model no.	Leaving Chilled water tempaure (C)	Cooling Capacity (kW)	Component Capacity (kW)	COP	Water Flow (LPS)
ALZAMIL	ASQ220B	6	603.4	314.1	1.8	24.112250X422



Air Handling Unit (AHU)		
DAIKIN ZONE 1	1315	3900mm
DAIKIN ZONE 2	1010	3700mm
DAIKIN ZONE 3	1010	3700mm
DAIKIN ZONE 4	1013	3700mm



Fan coil unit		
YORK	940	530x1415x255
YORK	740	530x1415x255
YORK	540	530x1200x255

YFCN Fan Coil Unit with centrifugal fan
1.0 to 7.6 kW

Technical features

Model	max	240	240	340	440	540	640	740	840	940
Total cooling capacity (kW)	(1) max	1.02	1.41	1.89	2.28	3.25	3.88	4.64	5.71	6.54
	(2) med	0.67	1.01	1.65	2.83	2.19	2.83	3.96	4.08	4.88

Diffuser		
ADA	ADCD	600X600
ADA	ADCD	300x300

CEILING DIFFUSER

Features:

- Industry standard louvre faced ceiling supply air grille
- 75mm flush frame with removable core
- Aluminium lightweight construction
- Easy to install
- Core removable for access to dampers
- Powder coated finish

ADCD

Duct Sizing Table					
S.no	Section	Length	Airflow (L/s)	Velocity (m/s)	Size (mm x mm)
1	A-B	4.8	127	3	100X300
2	B-C	1.3	254	3	200X400
3	C-F	3	381	3	200X500
1	D-E	1.5	127	10	200X300
2	E-F	1.8	254	10	200X400
5	F-G	1.6	762	10	200X400
6	G-H	3	889	10	200X400
7	H-I	3	1016	10	200X500
8	I-J	1.1	1143	10	200X600
1	K-J	6.2	127	3	100X300
9	J-L	1.8	1143	10	200X600
10	L-M	3	1270	10	200X600
11	M-N	3	1397	10	200X800
12	N-S	1.2	1524	10	200X800
1	O-P	3.5	127	3	100X300
2	P-Q	4	254	3	200X400
3	Q-R	4.6	381	3	200X500
4	R-S	4	508	3	200X800
16	S-T	1.1	2032	10	250X800
17	T-V	1.9	2159	10	250X800
1	U-V	6.5	127	3	100X300
35	V-W	1.1	4445	10	600X800
36	W-X	3	4572	10	600X800
37	X-Z	5	4699	10	600X800
1	A2-B2	2.6	127	3	100X300
2	B2-C2	3.5	254	3	200X400
3	C2-G2	0.9	381	3	200X500
1	D2-E2	2.9	127	3	100X300
2	E2-F2	3.7	254	3	200X400
3	F2-G2	1	381	3	200X500
6	G2-I2	4.4	762	10	200X400
1	H2-I2	2.9	127	10	100X300
7	I2-J2	2.9	889	10	200X500
8	J2-P2	0.6	1016	10	200X600
1	L2-M2	4.3	127	3	100X300
2	M2-O2	2	254	3	200X400
1	N2-O2	2	127	3	100X300
3	O2-P2	4.7	381	3	200X500
11	P2-U2	5.1	1397	10	200X800
12	U2-V2	1.8	1524	10	200X800
1	Q2-R2	4.3	127	3	100X300
2	R2-T2	2	254	3	200X400
1	S2-T2	2	127	3	100X300
3	T2-V2	4.7	381	3	200X500
15	V2-W2	4.1	1905	10	250X800
16	W2-X2	6	2032	10	250X800
17	X2-V	11	2159	10	250X800

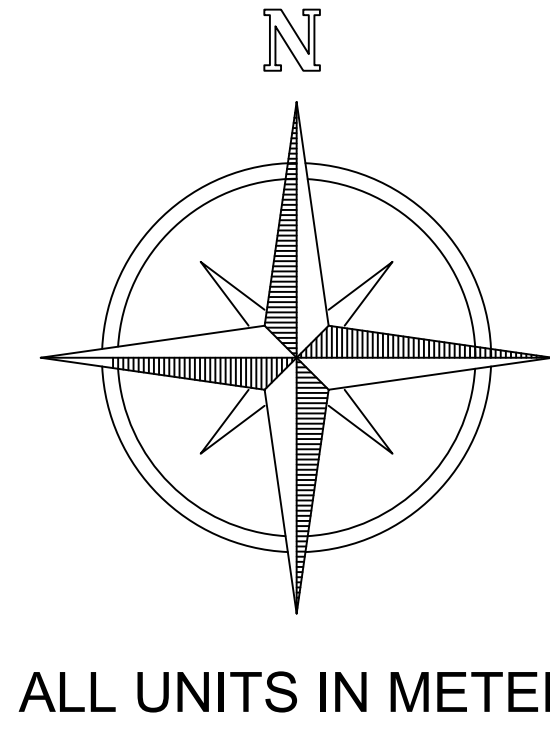
CALCULATION (Peak Cooling Load, Quantity of Air, No and Capacity of Supply/Return Diffusers etc.)													
S. No.	Zones/Space	Type of HVAC system	QYT	Area (m²)	Height (m)	Volume (m³)	Peak Load (Ton of Refrigeration)	Peak load (kW)	Quantity of Air (CFM)	Quantity of Air (L/s)	No of Supply Diffuser	No of Return Diffuser	Capacity of each duffuser (L/s)
Ground Floor													
1	Corridors (Zone 1)	All Air system - Variable Air Volume (VAV)	1	228	2.7	615.6	10.3	36.1	4104	1929	15	15	127
2	Cafe (Zone 1)	All Air system - Variable Air Volume (VAV)	1	61	2.7	164.7	2.7	9.7	1098	516	4	4	127
3	Storage (Zone 1)	All Air system - Variable Air Volume (VAV)	1	19	2.7	51.3	0.9	3.0	342	161	1	1	127
4	Offices (Zone 1)	All Air system - Variable Air Volume (VAV)	1	41	2.7	110.7	1.8	6.5	738	347	3	3	127
5	Store (Zone 1)	All Air system - Variable Air Volume (VAV)	1	41	2.7	110.7	1.8	6.5	738	347	3	3	127
6	Barber shop (Zone 1)	All Air system - Variable Air Volume (VAV)	1	41	2.7	110.7	1.8	6.5	738	347	3	3	127
7	electric room (Zone 1)	All Air system - Variable Air Volume (VAV)	1	20	2.7	54	0.9	3.2	360	169	1	1	127
8	Dry cleaner (Zone 1)	All Air system - Variable Air Volume (VAV)	1	88	2.7	237.6	4.0	13.9	1584	744	6	6	127
1	Comidors (Zone 2)	All Air system - Variable Air Volume (VAV)	1	243	2.7	656.1	10.9	38.4	4374	2056	16	16	127
2	electric (Zone 2)	All Air system - Variable Air Volume (VAV)	1	20	2.7	54	0.9	3.2	360	169	1	1	127
3	Toilets (Zone 2)	All Air system - Variable Air Volume (VAV)	1	30	2.7	81	1.4	4.7	540	254	2	2	127
4	study room (Zone 2)	All Air system - Variable Air Volume (VAV)	1	61	2.7	164.7	2.7	9.7	1098	516	4	4	127
5	Dry cleaner (Zone 1)	All Air system - Variable Air Volume (VAV)	1	88	2.7	237.6	4.0	13.9	1584	744	6	6	127
1	HC BedRoom1	All water system/ HC BedRoom1	1	40.6	2.7	109.62	1.8	6.4	731	343	3	3	127
2	HC BedRoom 2	All water system/ HC BedRoom2	1	40.6	2.7	109.62	1.8	6.4	731	343	3	3	127
1	BedRoom 1	All water system/ Bed Room 1	1	20	2.7	54	0.9	3.2	360	169	1	1	127
2	BedRoom 2	All water system/ Bed Room 2	1	20	2.7	54	0.9	3.2	360	169	1	1	127
3	BedRoom 3	All water system/ Bed Room 3	1	20	2.7	54	0.9	3.2	360	169	1	1	127
4	BedRoom 4	All water system/ Bed Room 4	1	20	2.7	54	0.9	3.2	360	169	1	1	127
5	BedRoom 5	All water system/ Bed Room 5	1	20	2.7	54	0.9	3.2	360	169	1	1	127
6	BedRoom 6	All water system/ Bed Room 6	1	20	2.7	54	0.9	3.2	360	169	1	1	127
7	BedRoom 7	All water system/ Bed Room 7	1	20	2.7	54	0.9	3.2	360	169	1	1	127
8	BedRoom 8	All water system/ Bed Room 8	1	20	2.7	54	0.9	3.2	360	169	1	1	127
9	BedRoom 9	All water system/ Bed Room 9	1	20	2.7	54	0.9	3.2	360	169	1	1	127
10	BedRoom 10	All water system/ Bed Room 10	1	20	2.7	54	0.9	3.2	360	169	1	1	127
11	BedRoom 11	All water system/ Bed Room 11	1	20	2.7	54	0.9	3.2	360	169	1	1	127
12	BedRoom 12	All water system/ Bed Room 12	1	20	2.7	54	0.9	3.2	360	169	1	1	127
13	BedRoom 13	All water system/ Bed Room 13	1	20	2.7	54	0.9	3.2	360	169	1	1	127
14	BedRoom 14	All water system/ Bed Room 14	1	20	2.7	54	0.9	3.2	360	169	1	1	127
1	Large BedRoom 1	All water system/ Large Bed Room 1	1	30	2.7	81	1.4	4.7	540	254	2	2	127
2	Large BedRoom 2	All water system/ Large Bed Room 2	1	30	2.7	81	1.4	4.7	540	254	2	2	127
3	Large BedRoom 3	All water system/ Large Bed Room 3	1	30	2.7	81	1.4	4.7	540	254	2	2	127
4	Large BedRoom 4	All water system/ Large Bed Room 4	1	30	2.7	81	1.4	4.7	540	254	2	2	127
5	Large BedRoom 5	All water system/ Large Bed Room 5	1	30	2.7	81	1.4	4.7	540	254	2	2	127
6	Large BedRoom 6	All water system/ Large Bed Room 6	1	30	2.7	81	1.4	4.7	540	254	2	2	127
7	Large BedRoom 7	All water system/ Large Bed Room 7	1	30	2.7	81	1.4	4.7	540	254	2	2	127
8	Large BedRoom 8	All water system/ Large Bed Room 8	1	30	2.7	81	1.4	4.7	540	254	2	2	127
9	Large BedRoom 9	All water system/ Large Bed Room 9	1	30	2.7	81	1.4	4.7	540	254	2	2	127
Sub Total				1612.2			72.549	255.1	29019.6	13639.212			
GRAND TOTAL							139.959	492.1	55983.6	26312.292			

TOTAL AIR QUANTITY FOR ZONE (1) TO FIND AHU

#N. OF DIFFUSERS & THEIR CAPACITY
#O. FOR ZONE (1) TO FIND DUCT SIZES

AIR QUANTITY FOR ROOM TO FIND FCU

PEAK TOTAL LOAD TO FIND CHILLER



CHILLER FROM:
Cooline
AIR CONDITIONERS
Zamil

AHU FROM:
DAIKIN

FCU FROM:
YORK
BY JOHNSON CONTROLS

DIFFUSERS FROM:
ADA
AIR DIFFUSION AGENTS

Design Studio (6), Technical
Project, 2023/2024

INSTRUCTORS:
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DONE BY:
ABDULELAH ALQABBANI

PROJECT:
STUDENT DORM

DRAWING NAME:
HVAC CALCULATIONS
& EQUIPMENT

SCALE:
1:100

SHEET NUMBER:
A16